

Matrix functions and stability

# Local Hölder continuity of the joint spectral radius

**Difficulty:** challenging **Importance:** interesting to the community**Status:** Solved

**Rating rationale:** Challenging because reducible families obstruct uniform perturbation estimates; community impact comes from conditioning and robustness of stability computations.

**Resolution — 2026-09-11**

**Affirmative resolution.** Matthew J. Colbrook's [complete manuscript](#), [Theorem 2](#) and [Corollary 7](#) proves the stronger uniform estimate

$$|\hat{\rho}(\mathcal{M}) - \hat{\rho}(\mathcal{N})| \leq d(2d+1)L^{1-1/d}d_H(\mathcal{M}, \mathcal{N})^{1/d}$$

for any two nonempty compact real or complex matrix families in the spectral-norm ball of radius  $L > 0$ . Choosing a common norm ball around a fixed family gives the exact local two-family assertion below, including reducible families and zero joint spectral radius. For  $d = 1$  the Lipschitz constant is one. The exponent  $1/d$  is sharp in general; no Lipschitz lower-bound resolution of MF-06 is asserted.

The complete original proof passed [independent Codex-agent review](#). [Authored TeX · Submission, authorship and verification record](#). The proof was developed with AI assistance; no external human peer review or formal verification is claimed. The original statement and prior evidence below are retained, and the ratings above are historical. This entry no longer contributes to the open count.

## Context and notation

Let  $\mathcal{H}_d$  denote the nonempty compact subsets of  $\mathbb{C}^{d \times d}$ . Use the spectral norm and its Hausdorff distance

$$d_H(\mathcal{M}, \mathcal{N}) = \max \left\{ \sup_{A \in \mathcal{M}} \inf_{B \in \mathcal{N}} \|A - B\|_2, \sup_{B \in \mathcal{N}} \inf_{A \in \mathcal{M}} \|A - B\|_2 \right\}.$$

The joint spectral radius is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

These definitions also apply to finite real matrix sets. The ordinary spectral radius of one matrix is written  $\rho(A)$ .

## Problem statement

For every  $d \geq 1$  and  $\mathcal{M}_0 \in \mathcal{H}_d$ , do there exist  $r, C > 0$  such that

$$|\hat{\rho}(\mathcal{M}) - \hat{\rho}(\mathcal{N})| \leq Cd_H(\mathcal{M}, \mathcal{N})^{1/d}$$

whenever  $d_H(\mathcal{M}, \mathcal{M}_0) < r$  and  $d_H(\mathcal{N}, \mathcal{M}_0) < r$ ?

## Reference and status evidence

Epperlein and Wirth, [The joint spectral radius is pointwise Hölder continuous](#), Linear Algebra Appl. 704 (2025), 92–122, [DOI](#), §2, Conjecture 3 (L1). Pointwise results in that paper do not prove this local assertion.

**Additional status evidence**

Searches combining “joint spectral radius” with “local Hölder”, “Lipschitz lower”, “trajectory bounds”, the authors’ names, and 2025/2026 found no later resolution. These searches supplement the explicit 2025 conjectures; they do not establish exhaustiveness. The three entries are separately named assertions in the source, not a count of dimensional cases.

**Audit — 2026-09-10**

Rechecked [Epperlein–Wirth, §1 and Conjecture 3 \(L1\)](#). Local Lipschitz continuity settles the irreducible-family subcase, but the displayed exponent for arbitrary compact families remains open. Author and local-Hölder follow-up searches found no resolution; the weaker general pointwise results do not suffice.